Pras Bairwa

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Computer Science @ UIUC (GPA: 4.0) | AI / Software Engineering Intern

Experience

Kongsberg Digital, Oslo, Norway
Artificial Intelligence Intern — *Jun. 2025 – Aug. 2025*

- **Designed and implemented a Retrieval-Augmented Generation (RAG)** system, reducing asset data query latency by **70%** by optimizing embedding storage, retrieval strategy, and prompt orchestration for real-time industrial decision support.
- **Architected agentic AI pipelines** using LangChain/LangGraph with Reflexion and CRAG patterns to convert unstructured technical documentation into structured, queryable knowledge graphs.
- **Integrated multimodal RAG pipelines** (text + logs) using **Docker** for reproducible workflows and **Azure DevOps CI/CD** to support testing, evaluation, and reliability.

Cellular Neuroscience Imaging Laboratory, UIUC
Research Assistant — *Aug. 2024 – Present*

- **Developed an end-to-end computer vision pipeline** (CNNs, YOLO, Cellpose) for label-free SLIM microscopy segmentation, enabling **high-throughput analysis** without manual annotations.
- **Prototyped deep learning classifiers** for collagen pattern recognition in breast tissue, contributing to **early-stage cancer diagnostic research**.

CS233: Computer Architecture, UIUC
Course Assistant — *Jan. 2026 – Present*

- **Contributed to backend development and maintenance** of SpimBot, a MIPS based autonomous agent simulator.
- Improved robustness and maintainability of the simulation **backend used by 200+ students**.
- Assisted students in understanding instruction execution, pipeline behavior, control flow, and memory hierarchy concepts.

Projects

- **Transformer Implementation (PyTorch)**: **Implemented an encoder–decoder Transformer from scratch in PyTorch**, including custom multi-head attention and positional encoding, to deepen understanding of sequence modeling internals.
- **Local RAG System**: Implemented a RAG pipeline using ChromaDB, Ollama embeddings, and LangChain text-splitting, converting unstructured documents into queryable embeddings and generating AI responses.
- **Federated Facial Detection (Rust)**: **Designed a privacy-preserving federated learning system in Rust**, implementing federated averaging across distributed clients to enable secure face detection without centralized data collection.

Technical Skills

Languages & Hardware: SQL, Python, Java, C++, C, Rust, Verilog, MIPS Assembly
Machine Learning: Deep Learning, Computer Vision, NLP, RAG, Multimodal AI, Federated Learning
Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenCV, Docker, Git, Linux, Azure DevOps

Education

University of Illinois Urbana-Champaign
B.S. in Computer Science, GPA: 4.0/4.0 — *2024 – 2027*
On accelerated track to graduate early due to advanced coursework and IB credits.

International School of Stavanger, Norway
International Baccalaureate 44/45; Salutatorian; 8 A IGCSEs* — *2020 – 2024*

Stanford University
Stanford University Mathematics Camp (Algebraic Topology) — *Summer 2023*

Awards & Honors

- **URSA Best Overall Award, UIUC (2024):** Recognized for top research presentation on label-free cellular segmentation, demonstrating excellence in deep learning and computer vision applications.
- **Illinois Engineering Achievement Scholar:** Awarded for academic excellence, research leadership, and innovation.
- **James Scholar, Phi Eta Sigma & Dean's List:** Maintained a perfect 4.0 GPA while undertaking advanced coursework and research.